	Design			Composition		Networks			Statistical information			Numerical conditioning			
Scenario	Cells per bulk n_i	Genes G	Cell-size ratio	Shannon $H^\star$	$\|\boldsymbol{p}\|_2^2$	Edge density	Mean degree	Hoyer sparsity	Mean overlap $\bar{\omega}$	Hellinger	max f_{cov}	$\kappa(\boldsymbol{\mu})$	$\lambda_{\min}(\boldsymbol{\mu}^\top\boldsymbol{\mu})$	$\kappa\{\boldsymbol{\Sigma}(\boldsymbol{p})\}$	$\lambda_{\min}\{\boldsymbol{\Sigma}(\boldsymbol{p})\}$
G = 20 / 1000 cells per bulk / balanced / low sibling overlap															
a = 5 / equal sizes	1e+03	20	1	1	0.25	0.1	1.9	0.52	0.00304	0.994	0.121	3.46	0.0167	4.3	0.000213
a = 5 / sizes 1:1:5:5	1e+03	20	5	1	0.25	0.1	1.9	0.611	0.00312	0.994	0.223	15	0.0167	5.71	0.0025
G = 20 / 1000 cells per bulk / balanced / high sibling overlap															
a = 5 / equal sizes	1e+03	20	1	1	0.25	0.1	1.9	0.52	0.0262	0.962	0.252	3.46	0.0167	4.3	0.00052
a = 5 / sizes 1:1:5:5	1e+03	20	5	1	0.25	0.1	1.9	0.611	0.0264	0.963	0.411	15	0.0167	5.71	0.00608
G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap															
a = 5 / equal sizes	1e+03	20	1	0.861	0.34	0.1	1.9	0.438	0.00412	0.994	0.413	3.46	0.0167	5.2	0.000284
a = 5 / sizes 1:1:5:5	1e+03	20	5	0.861	0.34	0.1	1.9	0.43	0.00479	0.994	0.372	15	0.0167	7.23	0.00311
G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap															
a = 5 / equal sizes	1e+03	20	1	0.861	0.34	0.1	1.9	0.438	0.0349	0.962	0.631	3.46	0.0167	5.2	0.000691
a = 5 / sizes 1:1:5:5	1e+03	20	5	0.861	0.34	0.1	1.9	0.43	0.0342	0.963	0.59	15	0.0167	7.23	0.00756
G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap															
a = 1 / equal sizes	1e+05	20	1	1	0.25	0.1	1.9	0.52	0.00325	0.994	0.121	3.46	0.0167	4.3	0.000213
a = 5 / equal sizes	1e+05	20	1	1	0.25	0.1	1.9	0.52	0.003	0.994	0.121	3.46	0.0167	4.3	0.000213
a = 5 / sizes 1:1:5:5	1e+05	20	5	1	0.25	0.1	1.9	0.611	0.00317	0.994	0.223	15	0.0167	5.71	0.0025
G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap															
a = 5 / equal sizes	1e+05	20	1	1	0.25	0.1	1.9	0.52	0.0262	0.962	0.252	3.46	0.0167	4.3	0.00052
a = 5 / sizes 1:1:5:5	1e+05	20	5	1	0.25	0.1	1.9	0.611	0.0265	0.963	0.411	15	0.0167	5.71	0.00608
G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap															
a = 5 / equal sizes	1e+05	20	1	0.861	0.34	0.1	1.9	0.438	0.00446	0.994	0.413	3.46	0.0167	5.2	0.000284
a = 5 / sizes 1:1:5:5	1e+05	20	5	0.861	0.34	0.1	1.9	0.43	0.00425	0.994	0.372	15	0.0167	7.23	0.00311
G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap															
a = 5 / equal sizes	1e+05	20	1	0.861	0.34	0.1	1.9	0.438	0.0352	0.962	0.631	3.46	0.0167	5.2	0.000691
a = 5 / sizes 1:1:5:5	1e+05	20	5	0.861	0.34	0.1	1.9	0.43	0.0342	0.963	0.59	15	0.0167	7.23	0.00756
G = 200 / 1000 cells per bulk / balanced / low sibling overlap															
a = 5 / equal sizes	1e+03	200	1	1	0.25	0.0499	9.94	0.419	0.00325	0.995	0.523	3.46	0.00167	7.92	2.63e-05
a = 5 / sizes 1:1:5:5	1e+03	200	5	1	0.25	0.0499	9.94	0.76	0.00329	0.995	0.821	15	0.00167	10.1	0.000361
G = 200 / 1000 cells per bulk / balanced / high sibling overlap															
a = 5 / equal sizes	1e+03	200	1	1	0.25	0.0499	9.94	0.419	0.0273	0.965	0.847	3.46	0.00167	7.92	0.000133
a = 5 / sizes 1:1:5:5	1e+03	200	5	1	0.25	0.0499	9.94	0.76	0.0272	0.965	0.959	15	0.00167	10.1	0.00182
G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap															
a = 5 / equal sizes	1e+03	200	1	0.861	0.34	0.0499	9.94	0.398	0.00429	0.995	0.89	3.46	0.00167	9.04	3.5e-05
a = 5 / sizes 1:1:5:5	1e+03	200	5	0.861	0.34	0.0499	9.94	0.712	0.00375	0.995	0.921	15	0.00167	12.8	0.000473
G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap															
a = 5 / equal sizes	1e+03	200	1	0.861	0.34	0.0499	9.94	0.398	0.0348	0.965	0.976	3.46	0.00167	9.04	0.000177
a = 5 / sizes 1:1:5:5	1e+03	200	5	0.861	0.34	0.0499	9.94	0.712	0.0342	0.965	0.983	15	0.00167	12.8	0.00239
G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap															
a = 5 / equal sizes	1e+05	200	1	1	0.25	0.0499	9.94	0.419	0.00308	0.995	0.523	3.46	0.00167	7.92	2.63e-05
a = 5 / sizes 1:1:5:5	1e+05	200	5	1	0.25	0.0499	9.94	0.76	0.00313	0.995	0.821	15	0.00167	10.1	0.000361
G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap															
a = 5 / equal sizes	1e+05	200	1	1	0.25	0.0499	9.94	0.419	0.0257	0.965	0.847	3.46	0.00167	7.92	0.000133
a = 5 / sizes 1:1:5:5	1e+05	200	5	1	0.25	0.0499	9.94	0.76	0.0272	0.965	0.959	15	0.00167	10.1	0.00182
G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap															
a = 5 / equal sizes	1e+05	200	1	0.861	0.34	0.0499	9.94	0.398	0.00371	0.995	0.89	3.46	0.00167	9.04	3.5e-05
a = 5 / sizes 1:1:5:5	1e+05	200	5	0.861	0.34	0.0499	9.94	0.712	0.00358	0.995	0.921	15	0.00167	12.8	0.000473
G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap															
a = 5 / equal sizes	1e+05	200	1	0.861	0.34	0.0499	9.94	0.398	0.0355	0.965	0.976	3.46	0.00167	9.04	0.000177
a = 5 / sizes 1:1:5:5	1e+05	200	5	0.861	0.34	0.0499	9.94	0.712	0.036	0.965	0.983	15	0.00167	12.8	0.00239